Marco Freschi

POSTDOCTORAL FELLOW IN MECHANICAL ENGINEERING

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RESEARCH INTERESTS

Functional and responsive materials, Bioinspired design, Materials for environmental and social impact

PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow

University of California, Berkeley (USA), 2023 - present

"*Bioinspired Design of Sustainable Seeding Methods for Improved Ecological Restoration*"

Advisor: Prof. Lining Yao

- Implemented the design of the bio-inspired seeding device, increasing the drilling success rate up to 75%.
- Tested the design at the lab-scale, simulating rain soil conditions, and real-world conditions, with deployment, drilling, and germination tests in the California forests and the Kazakh desert.
- Collaborated with partner US universities and an international Foundation, merging multidisciplinary contributions for the improvement of the seeding device design and testing.

Research Fellow

Politecnico di Milano (Italy), 2023

"*Sustainability strategies in the sector of materials for industrial packaging*"

Supervisor: Prof. Giovanni Dotelli - *in collaboration with BCube*

EDUCATION AND TRAINING

PhD

Politecnico di Milano (Italy), 2022

Materials Engineering

"Production and characterization of self-lubricating metal matrix composites for sliding electrical contact in aerospace applications"

Supervisor: Prof. Giovanni Dotelli - *in collaboration with Logic S.p.A.*

Visiting period at Oulu University (Finland) - Supervisor: PhD O. Haiko, Prof. J. Kömi

- Designed and developed the composite material via powder metallurgy.
- Characterized the composite material through a comprehensive multiscale approach, from powder morphology to surface and tribological properties, using spectroscopic, microscopic, and mechanical techniques.
- Identified the best composition of copper and micro- and nano- tungsten disulfide solid lubricant for the application, satisfying the requirements of the collaborating company.

Master's degree

Politecnico di Milano (Italy), 2016

Materials Engineering and Nanotechnology

"*Experimental investigation of the air path in timber frame wall assembly*"

Internship at Université Savoie Mont Blanc – Chambéry (France), resulting in a peer-reviewed publication

Bachelor's degree

Politecnico di Milano (Italy), 2013

Materials Engineering and Nanotechnology

"*Anodization of titanium alloy – effect of the operative conditions on the optical and corrosion properties*"

PUBLICATIONS

In preparation and under review:

Two papers related to the project at the University of California, Berkeley, within the research group of Professor Yao, about the autonomous seed carrier technology and the project methodology.
One paper related to a Life Cycle Assessment study in collaboration with Politecnico di Milano, with the research group of Professor Dotelli.

Published (peer-reviewed):

- M. Freschi, M. Di Virgilio, O. Haiko, M. Mariani, L. Andena, N. Lecis, J. Kömi, G. Dotelli
  "*Analysis of ball milling time to produce self-lubricating copper-tungsten disulfide composite: Best trade-off between tribological performance and electrical properties*"
  Next Materials, 2024, 2, 100083. DOI: 10.1016/j.nxmate.2023.100083
- M. Freschi, L. Dragoni, M. Mariani, O. Haiko, J. Kömi, N. Lecis, G. Dotelli
  "*Tuning the Parameters of Cu-WS_2 Composite Production via Powder Metallurgy: evaluation of the effects on tribological properties*"
  Lubricants, 2023, 11(2), 66. DOI: 10.3390/lubricants11020066
- M. Freschi, A. Paniz, E. Cerqueni, G. Colella, G. Dotelli
  "*The twelve principles of green tribology: studies, research, and case studies-a brief anthology*"
  Lubricants, 2022, 10, 129. DOI: 10.3390/lubricants10060129
- M. Freschi, A. Arrigoni, O. Haiko, L. Andena, J. Kömi, C. Castiglioni, G. Dotelli
  "*Physico-Mechanical Properties of Metal Matrix Self-Lubricating Composites Reinforced with Traditional and Nanometric Particles*" Lubricants, 2022, 10, 35. DOI: 10.3390/lubricants10030035
- M. Freschi, M. Di Virgilio, O. Haiko, M. Mariani, L. Andena, N. Lecis, J. Kömi, G. Dotelli
  "*Investigation of second phase concentration effects on tribological and electrical properties of Cu–WS_2 composites*"
  Tribology International, 2022, 166 (107357). DOI: 10.1016/j.triboint.2021.107357
- M. Freschi, M. Di Virgilio, G. Zanardi, M. Mariani, N. Lecis, G. Dotelli
  "*Employment of Micro- and Nano- WS_2 Structures to enhance the tribological properties of copper matrix composites*"
  Lubricants, 2021, 9(5), 53. DOI: 10.3390/lubricants9050053
- N. Hurel, M. Freschi, M. Pailha, M. Woloszyn
  “Innovative use of fluorescein for the air path study within light-weight wall assemblies”
  Experimental Thermal and Fluid Science, 2018, 92. DOI: 10.1016/j.expthermflusci.2017.10.007

MENTORING AND TEACHING

Total students supervised: 42

Undergraduate and graduate, UC Berkeley and Politecnico di Milano [2018-present]

University of California, Berkeley – co-supervisor for

Mechanical Engineering – 4 undergraduate students (student assistant) [2025]

Mechanical Engineering – 5 graduate students (capstone project) [2025-2026]

Electrical Engineering and Computer Sciences – 4 undergraduate students (final project class) [2025]

Politecnico di Milano – co-supervisor for thesis projects in

Materials and Chemical Engineering – 7 master's and 23 bachelor's students [2018-2022]

Politecnico di Milano – teaching assistant for the courses

School of Architecture (approximately 70 students per year)

Sustainable Materials for Architecture [2021-2022]

Environmental Sustainability and LCA of Building Materials [2020-2021]

Innovative Materials for Architecture [2019-2021]

Science and Technology of Materials [2018-2019]

School of Design (approximately 50 students per year)

Product Performance in Fashion Design [2018-2022]

SERVICE AND LEADERSHIP

2026	**Chancellor's Outstanding Staff Award** – Nominated staff member selected as recipient (University of California, Berkeley)
2026 – present	**Berkeley Postdoc Association – Community and Culture Committee Chair** (University of California, Berkeley)
2024 – present	**Berkeley Postdoc Association – Social Committee Co-chair** (University of California, Berkeley)
2023	**ERC Consolidator Grant 2023 Call Reviewer**
2023 – present	**Topical Advisor Panel member – Lubricants MDPI**
2019 – present	**Peer Reviewer** – Applied Sciences, Coatings, Journal of Bio- and Tribo-Corrosion, Lubricants, Machines, Materials Circular Economy, Polymers, RSC Advances, Sustainability, Tribology Letters, Wear
2021 – 2022	**Research Fellows Representative** – Department of Chemistry, Materials and Chemical Engineering Giulio Natta (Politecnico di Milano)
2018 – 2021	**PhD School Representative** (Politecnico di Milano)

CONFERENCES

Tribology 2022 – Barcelona (Spain), 2022
Oral: *Investigation of experimental production parameters effects on the wear behavior of copper-tungsten disulfide composite*

3rd PCNS–Passive Components Networking Symposium – Milano (Italy), 2021
Organizing Committee Member

AIMAT 2021 – XVI Convegno Nazionale – Cagliari (Italy), 2021
Poster: *Effect of second phase concentration on tribological properties of metal matrix self-lubricating composite materials*

K-Trib 2020: 2nd Korea-Tribology International Symposium – Seoul, 2020
Oral: *Employment of micro- and nano-WS_2 structures to enhance the tribological properties of copper-matrix composites*

Materials in the next decade – Favignana (Italy), 2019
Oral: *How to extend the lifetime of electrical sliding contacts through the material design*

Young Materials and Surface Engineers – Roma (Italy), 2019
Poster: *Analysis of the dependence of tribological performances on the properties of the pristine materials*

FUNDED PROJECTS AND FELLOWSHIPS

Bulat Utemuratov Foundation "Scientific Sponsorship Project" – PI: L. Yao – 2026 (Postdoctoral researcher)

Berkeley Climate Action Proof-of-Concept Program – PI: L. Yao – 2025 (Postdoctoral researcher, contributed to proposal writing)

Harvard Faculty of Art and Science Research and Academic Exchange – Recipient: M. Freschi – 2025

Enel Foundation Accelerator Postdoctoral Fellowship – Recipient: M. Freschi – 2023-2025

European Institute of Innovation and Technology "Raw Matters Ambassadors at Schools" – PI: G. Dotelli – 2016-2021 (PhD student)

European Institute of Innovation and Technology "ADMA2 – Practical Training Between Academia and Industry During Doctoral Studies" – PI: G. Dotelli – 2018-2021 (PhD student)

MEDIA AND OUTREACH

1. World Forestry Center Discovery Museum, Portland (OR), April 2026 – *Forest Hope Through Innovation*
2. CNN Tech for good, December 2025 – *These self-burying 'E-seeds' could be the future of reforestation*
3. Corriere della Sera, Italy Oct. 2025 – *I semi che si piantano da soli di Berkeley: così la riforestazione può cambiare marcia* (The self-burying seeds from Berkeley: how reforestation could take a leap forward)
4. UC Berkeley, Oct. 2025 – *Engineering Nature: Dr. Marco Freschi's Work on Self-Burying Seed Carriers*
5. Victoria and Albert Museum Dundee, Scotland, May 2025 – *Garden Futures: Designing with Nature*

TECHNICAL SKILLS

Fabrication: powder metallurgy, ball milling, sintering, 3D-printing, laser cutting

Characterization: SEM, XRD, Raman spectroscopy, Thermogravimetric analysis, Confocal laser scanning microscopy, Vickers micro-indentation, Tribometer (pin-on-disk, micro-scratch), Uniaxial tensile, compression, and bending testing, Custom environmental testing (rainfall simulation, soil penetration, germination assessment), Field testing (field deployment in California and Kazakhstan)

Software: Autodesk Inventor, Origin, ImageJ

Languages: Italian (native), English (fluent)